



**UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION**

COURSE	:	DATABASE DESIGN AND DEVELOPMENT
COURSE CODE	:	ICT450/ITS432
EXAMINATION	:	JANUARY 2024
TIME	:	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of FIVE (5) questions.
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
 - i. the Question Paper
 - ii. an Answer Booklet – provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 6 printed page

QUESTION 1

- a) Organizations use data to keep track of their day-to-day operations. Such data is likely to be managed most efficiently when it is stored in a database. Explain the meaning of data and database.
(4 marks)
- b) The Database Management System (DBMS) serves as the intermediary between the user and the database. One of the advantages of DBMS is minimized data inconsistency. Briefly explain data inconsistency and illustrate **ONE (1)** example.
(4 marks)
- c) Zinnirah has been appointed as a junior database developer in ZY Stationary company. She is tasked to create a database system to keep track of its customers, orders, and products. Zinnirah has created a database that consists of three tables which are Customers, Orders, and Products. However, the relational operations become very complex and are likely to lead to system efficiency errors and output errors. State the problem that Zinnirah is facing and illustrate the solution (an ERD) to overcome the problem.
(6 marks)
- d) Maintenance is one of the important phases in Systems Development Life Cycle (SDLC). Briefly explain **THREE (3)** types of maintenance in SDLC.
(6 marks)

QUESTION 2

- a) Conceptual model is the level of abstraction that defines the overall structure of database. Describe **TWO (2)** advantages of conceptual model for business environment.
(4 marks)
- b) The following Table 1 and Table 2 show the data from a publishing company. Briefly explain whether these two tables violate entity integrity and referential integrity. Justify your answer for each table.

Table 1: BOOKS table

ISBN	BookTitle	Authors	PublisherID
978-981-4824-24-7	Database Systems	Coronel, Morris	101
978-967-388-341-7	Innovation via Arduino	Hafizah Hussain	103
978-967-86-1017-9	Healing the Invisible Wound	Ahmad Rostam, Rozanizam, Alizi Alias	
978-981-4824-24-7	Digital Multimedia	Nigel, Jenny	102

Table 2: PUBLISHERS table

PublisherID	PublisherName	Country
101	Cengage Learning	Singapore
102	John Wiley & Sons	United States
103	Telaga Biru Sdn. Bhd.	Malaysia
104	Karangkraf Media Group	Malaysia

(4 marks)

- c) Figure 1 shows the procedure flow in the database design. Fill in the empty boxes with correct answers for the stages and hardware/software dependent for this database design process.

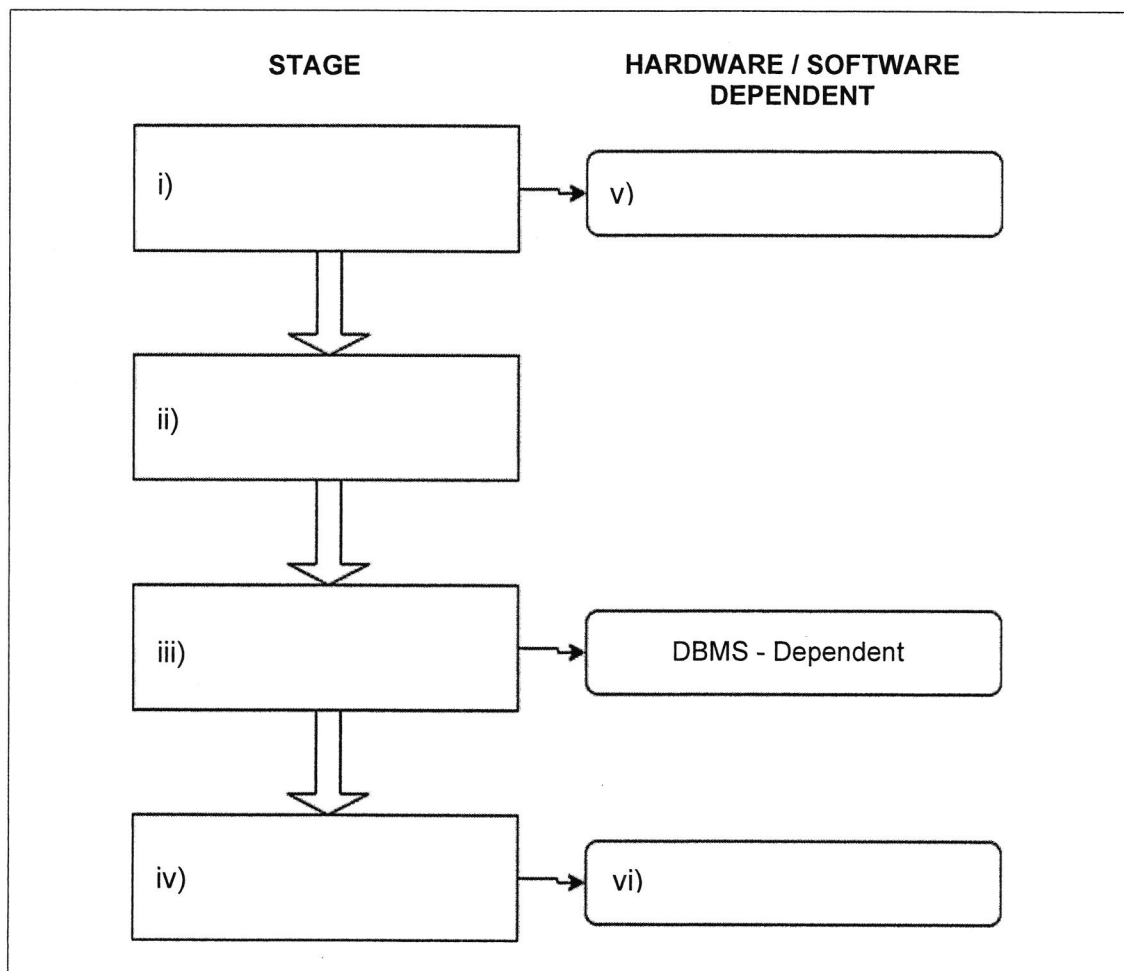


Figure 1 : Database Design Process

(6 marks)

- d) Describe **THREE (3)** factors that contribute to the increasing reliance on Big Data.

(6 marks)

QUESTION 3

Figure 2 shows a report of an employee evaluation result in IT Creative Solutions Sdn. Bhd. in Shah Alam. The report contains all information related to one of the employees in this company. Each employee has been assigned to several projects. Meanwhile, each project has been assigned to several employees. Each project has been supervised by one employee (Supervisor). Supervisor also manages several employees.

IT CREATIVE SOLUTIONS SDN. BHD. EMPLOYEE MONTHLY EVALUATION RESULT (FEBRUARY 2024)					
Employee ID : 303303					
Employee Name : Siti Nur Fadilah					
Employment Year : 2000					
Department : S1 - Software Development					
Phone Number : 017					
Permanent Address : JC 1234, Bandar Jasin Bestari Seksyen 7, Melaka					
Project Code	Project Name	Location	Grade	Supervisor ID & Name	Supervisor Department
MD01	Mobile Apps Development	Kuala Terengganu	A	123123 - Willy	Mobile Application Development
WD01	Website Development	Machang	A+	101101 - Mawar	Software Development
ST01	Software Integration	Seremban	A-	101101 - Mawar	Software Development
ST01	Software Integration	Tapah	A	101101 - Mawar	Software Development

Figure 2 : Employee evaluation report

Based on the information given, answer the following questions:

a) State the normal form of the Figure 2 above. Explain your answer with reasons.

(3 marks)

- b) Convert the report into a relational schema. Identify the primary key(s) and partial dependencies and/or transitive dependencies.
(6 marks)
- c) Normalize the relational schema in (b) until third normal form (3NF). Show all the steps involved.
(11 marks)

QUESTION 4

The relational schema below shows the movie database system.

MOVIE (<u>MovieID</u> , MovieName, Genre, DateRelease, NetProfit, DirectorID*)
DIRECTOR (<u>DirectorID</u> , DirectorName, Nationality)
REVIEW (<u>ReviewID</u> , Description, MovieID*, ReviewerID*)
REVIEWER (<u>ReviewerID</u> , ReviewerName)

Write the SQL statements based on the following requirements:

- a) Display the movies that have been released from 2021 until 2023.
(4 marks)
- b) Identify the total movies that have been directed by Malaysian directors.
(4 marks)
- c) Calculate the average net profit for each genre and name the output as "Average Net Profit".
(4 marks)
- d) Identify the reviewer's name(s) that has made more than 2 reviews.
(4 marks)
- e) Display the movie that received the most reviews.
(4 marks)

QUESTION 5

One public university in Arau, Perlis is planning to develop an auxiliary police database system for a better coordination and reporting of summons and security duties. As a database designer, you are tasked to design the database system according to the following requirements:

- Details of each police officer such as their Police ID, name, position and email address.
- Each police officer has been assigned to five main duties. For safety reasons, each duty will be carried out by more than one police officer. The date and time of each assigned duty need to be recorded in the system.
- Each police officer also can issue summons to several students. The date summon issuance need to be recorded by the police officer.
- A summon will be issued if a student violates any rules and regulations.
- Each summon will consist more than one offense. Offense ID, offense name, description, and rate of fees need to be recorded in the system.

Based on the given information:

- a) Construct a complete Entity Relationship Diagram (ERD) with all of the required tables, attributes, and relationships using the Crow's Foot notation that reflects all of the given business rules.

(16 marks)

- b) Identify **TWO (2)** relevant information that you can get from the ERD you constructed in (a).

(4 marks)

END OF QUESTION PAPER